Tutorial - 4

17/2/26

Logistic Regression

SW(x_1)	EL(x_2)	Attach (x_3)	Spam(y)	z	$\hat{y}$ Pred	$y - \hat{y}$
5	2	1	1	8	0.99	
1	0	0	0	1	0.78	
3	1	0	1	4	0.98	
0	0	0	0	0	0.5	
$\downarrow$ w_1	$\downarrow$ w_2	$\downarrow$ w_3				

Let $w_1 = 1$

$w_2 = 1$, $w_3 = 1$

$\boxed{\text{Bias}: w_0 = 0}$

first we have to calculate weight sum

$$\boxed{z = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3}$$

$z_1 = 0 + 1(5) + 1(2) + 1(1)$

$= 5 + 2 + 1$

$= 8$

$z_2 = 0 + 1(1) + 1(0) + 1(0)$

$= 1$

$z_3 = 4$

$z_4 = 0$

$\hat{y}_{pred} = \sigma(z)$

$$= \frac{1}{1+e^{-z}}$$

$$= \frac{1}{1+e^{-8}} = 0.99$$

$\Rightarrow 0.99 \geq 0.5 \Rightarrow$ so it is 1

$0.78 \geq 0.5 \Rightarrow$ so it is 1

$0.98 \geq 0.5 \Rightarrow 1$

$0.5 \geq 0.5 \Rightarrow 1$

$y - \hat{y}_{pred}$
0.01
-0.78
0.02
-0.5

$$CF = J = \frac{1}{N} \sum_{i=1}^{N} \left[y \log(\hat{y}) + (1-y) \log(1-\hat{y}) \right]$$

$$\frac{\partial J}{\partial w} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)$$

$$= \frac{1}{4} \left(0.01 + (-0.78) + 0.02 + (-0.5) \right)$$

$$\boxed{CF = -0.3125}$$

$$w_0^{new} = w_0^{old} + \eta \frac{1}{N} \sum_i (y_i - \hat{y}_i) \qquad \boxed{\eta = 1}$$

$$= 0 + (1) \frac{1}{4} (-0.3125)$$

$$= -0.078$$

Bias updating

<u>weights</u>:

$$w_j^{new} = w_j^{old} + \eta \frac{1}{N} \sum_i (y_i - \hat{y}_i) x_j$$

$$= w_j^{old} + \eta \left(\frac{\partial J}{\partial w}\right) x_j$$

$$w_1^{new} = 1 + \frac{1}{4}\{(0.01 \times 5) + (-0.78 \times 1) + (0.02 \times 3) + (-0.5 \times 0)$$

$$= 0.8325$$

$$w_2^{new} = 1 + \frac{1}{4}\left((0.01 \times 2) + (-0.78 \times 0) + (0.02 \times 1) + (-0.5 \times 0)\right)$$

$$= 1.01$$

$$w_3^{new} = 1 + \frac{1}{4}\left((0.01 \times 1) + (-0.78 \times 0) + (0.02 \times 0) + (-0.5 \times 0)\right)$$

$$= 1.0025$$

$$z_{New} = w_0^{New} + w_1^{New} x_1 + w_2^{New} x_2 + \ldots$$

$$= -0.3125 + 0.8325(5) + 1.01(2) + 1.0025(1)$$

$$z_1^{New} = 6.8725 \ (>0.5) \ (1)$$

$$z_2^{New} = -0.3125 = 0.52$$

$$z_3^{New} = 3.195 \ (1)$$

$$z_4^{New} = (0) = 0.3125$$